

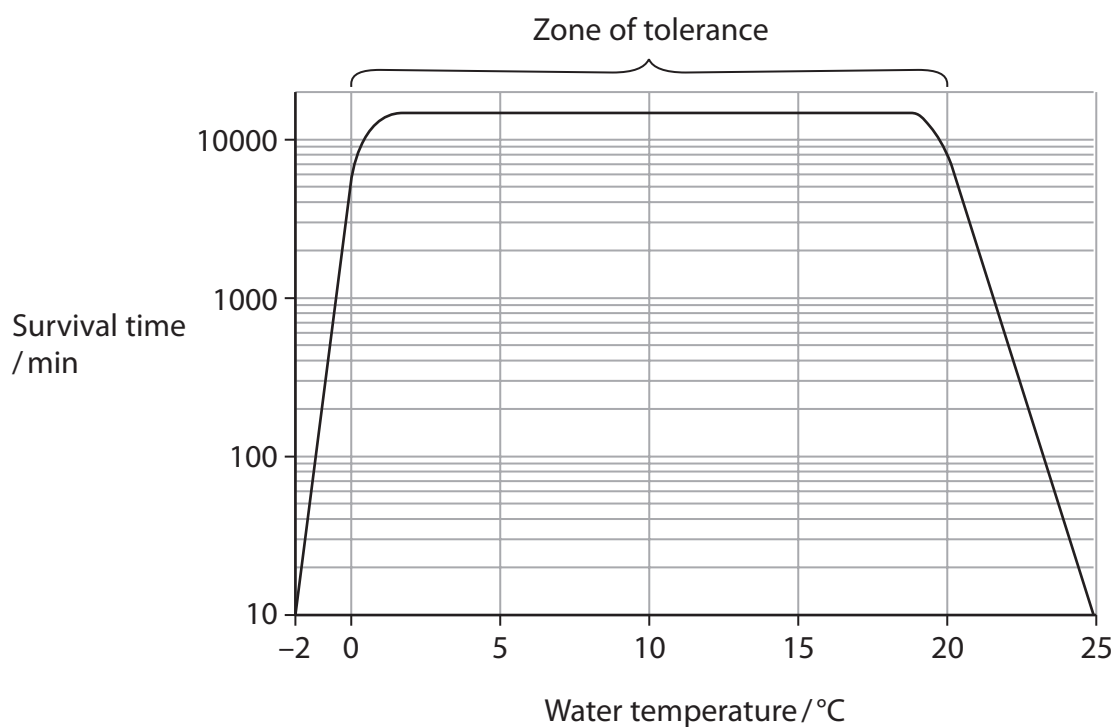
- 6 Changes in temperature affect the rate of enzyme activity and the survival of living organisms.

(a) Which describes an enzyme in a living organism?

(1)

- ☐ A inorganic substance that decreases activation energy
- ☐ B inorganic substance that increases activation energy
- ☐ C organic substance that decreases activation energy
- ☐ D organic substance that increases activation energy

(b) The graph shows the survival time of one species of fish in water at different temperatures.



(i) Name the type of scale that has been used on the y axis.

(1)

- (ii) The zone of tolerance is the range of temperatures that a fish could survive at for longer than 5000 minutes.

Which **two** biotic factors could cause the fish to die in this range of temperatures?

(1)

- ☐ **A** level of salt in the water and wave action
- ☐ **B** pathogens and wave action
- ☐ **C** predators and pathogens
- ☐ **D** predators and level of salt in the water

- (iii) The body temperature of a fish changes with the temperature of its surroundings.

Explain why the survival time of fish is reduced at temperatures lower and higher than those in the zone of tolerance.

(3)

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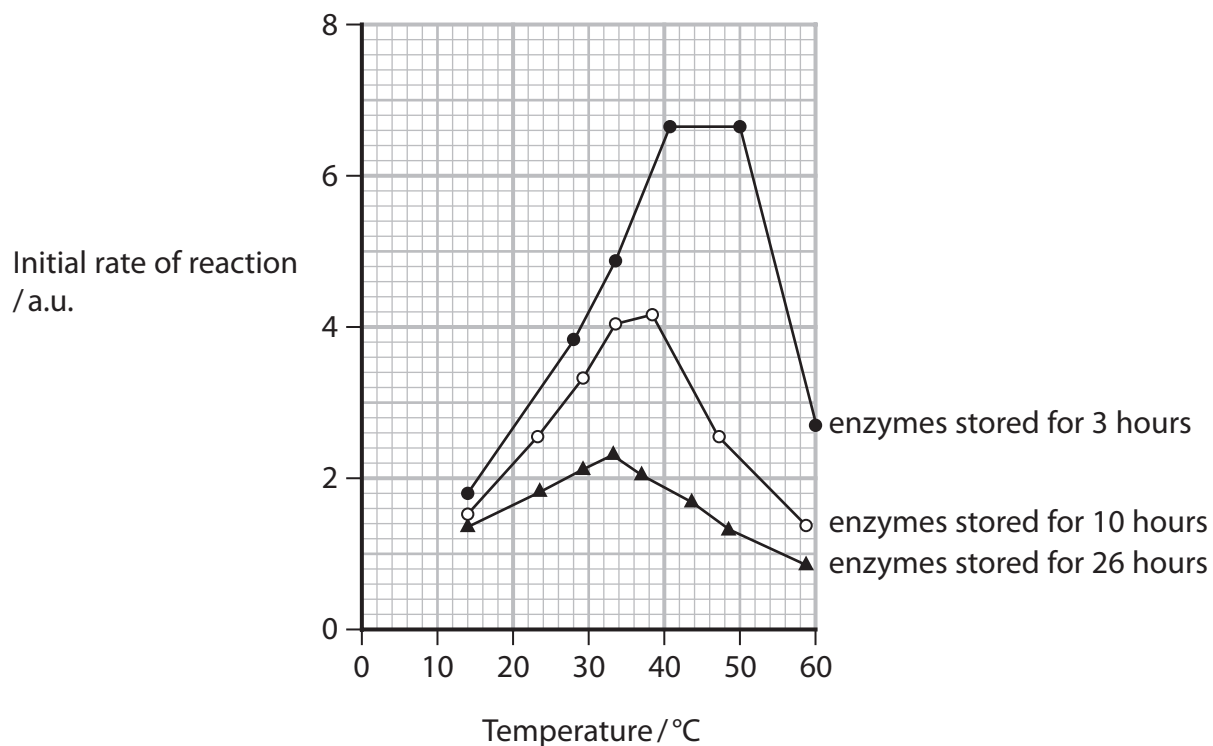
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- (c) Enzymes were stored for different lengths of time.
The effect on the initial rate of reaction was investigated.

The graph shows the results of this investigation.



- (i) Calculate the Q_{10} for this enzyme stored for 3 hours.

Use the formula

$$Q_{10} = \frac{R_{(t+10)}}{R_t}$$

where R_t is the initial rate of reaction at 20°C.

(3)

Answer.....

(ii) Explain why the optimum temperature for this enzyme is affected by storage time.

(3)

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(Total for Question 6 = 12 marks)

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